OBJECT ORIENTED PROGRAMMING
[CT-260]

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LAB: 1

CODES

```
//Q1. Write a program that take input of your roll number along with the marks obtained
//in five subjects and display the total marks obtained and the percentage.
#include <iostream>
using namespace std;
int main(){
  int rollno;
  float marks,percentage;
  cout<<"Enter your Rollno and Marks obtained in five subjects: ";
  cin>> rollno >>marks;
  cout<<"Total Marks in five subjects: "<<marks<<endl;
  percentage = (marks/500)*100;
  cout<<"percentage is: "<<percentage << "%";
}
//Q2. Write a program to swap three numbers entered by a user using pointers.

int main(){

  int a, b, c, temp;

  int *pqa = &a;
  int *pqb = &b;
  int *pqc = &c;

  cout<<"Enter the value of a b c: "<<endl;
  cin>>a>>b>>c;

  cout<<a<<" "<<b<<" "<<c<<endl;

  temp = *pqa;
  *pqa = *pqb;
  *pqb = *pqc;
  *pqc = temp;

  cout<<a<<" "<<b<<" "<<c<<endl;

  return 0;
}
```

```
//Q3. Write a program to convert temp from Fahrenheit to Celsius unit using equation
//C=(F-32)/1.8
int main(){
  double fahrenheit;
  double celsius;

  cout<<"Enter temp in fahrenheit: "<<endl;
  cin>>fahrenheit;

  celsius = (fahrenheit - 32) / 1.8;

  cout<<"The temp in celsius is "<<celsius<<endl;

  return 0;
}

//Q4. Using 2-D arrays, write a program that allows the user to input two, 3x3 matrices.
//Write a function for adding two matrices. Write another function for multiplying
//the two matrices.
#include<iostream>
using namespace std;

void sumMatrix(int arrA[3][3], int arrB[3][3], int sum[3][3]){
  int i,j;
  for(i=0; i<3; i++){
    for(j=0; j<3; j++){
      sum[i][j] = arrA[i][j] + arrB[i][j];

    }
  }
}
```

```
void multiplyMatrix(int arrA[3][3], int arrB[3][3], int product[3][3]){
  int i,j,k;

  for(i=0; i<3; i++){
    for(j=0; j<3; j++){
      product[i][j] = 0;
      for(k=0; k<3; k++){
        product[i][j] += arrA[i][k] * arrB[k][j];
      }
    }
  }
}

void display(int arr[3][3]){
  int i, j;
  for(i=0; i<3; i++){
    for(j=0; j<3; j++){
      cout<<arr[i][j]<<" ";
    }
    cout<<endl;
}
}

int main(){
  int arrA[3][3], arrB[3][3];
  int sum[3][3], product[3][3];
  int i,j,k;

  cout<<"Enter elements of first 3x3 matrix "<<endl;
  for(i=0; i<3; i++){
    for(j=0; j<3; j++){
      cin>>arrA[i][j];
    }
  }

  cout<<"Enter elements of second 3x3 matrix "<<endl;
  for(i=0; i<3; i++){
    for(j=0; j<3; j++){
      cin>>arrB[i][j];
    }
  }

  sumMatrix(arrA, arrB, sum);
  multiplyMatrix(arrA, arrB, product);

  cout<<"====Sum of Matrices===="<<endl;
  display(sum);

  cout<<"====Product of matrices===="<<endl;
  display(product);

  return 0;
}
```

```
//Q5. Write a program to find Surface area and volume of a sphere using functions.


float surfaceArea(float r){
  float surface;
  surface = 4 *3.14 *r *r;
  return surface;
}

float volSphere(float r){
  float vol;
  vol = (4.0/3.0) *(3.14) *r *r *r;
  return vol;
}

int main(){
  float r;
  cout<<"Enter the radius of sphere: "<<endl;
  cin>>r;

  surfaceArea(r);
  volSphere(r);

  cout<<"Surface Area of Sphere = "<<surfaceArea(r)<<endl;
  cout<<"Volume of Sphere = "<<volSphere(r)<<endl;

  return 0;
}
```

```
//Q6. Write a program to help a bank create its withdrawal system. Your program should
//allow the user to input their account type. Account types are: savings, current.
//Following business rules apply when withdrawing from a certain account:
//? Savings:
//User must provide the savings account number and code 'S' (for savings).
//When withdrawing from a savings account, users need to pay a set 2% of
//the money that they withdraw. If the amount of money withdrawn is over
//50,000, then a 5% tax will be deducted. The money deducted shall be from
//the remaining money in the account.
//? Current:
//User must provide the current account number and code „C" (for current).
//When withdrawing from a current account, users need to pay a withdrawal
//fee of 100. If the amount of money withdrawn is over 50,000, then a 5% tax
//will be deducted. The money deducted shall be from the remaining money
//in the account.
//Assume all users have the 200,000 in their accounts, and cannot withdraw more
//than 100,000 at a time.

int main() {
    char accountType;
    int accountNumber;
    float withdrawAmount;
    float balance = 200000;   // initial balance
    float deduction = 0;

    cout << "Enter account type (S for Savings, C for Current): ";
    cin >> accountType;

    cout << "Enter account number: ";
    cin >> accountNumber;

    cout << "Enter amount to withdraw: ";
    cin >> withdrawAmount;

    if (withdrawAmount > 100000) {
        cout << "Error: Cannot withdraw more than 100,000 at a time.";
        return 0;
```

```
        cout << "Error! Cannot withdraw more than 100,000 at a ti
        return 0;
    }

    if (accountType == 'S' || accountType == 's') {
        deduction = withdrawAmount * 0.02;   // 2% fee
        balance = balance - withdrawAmount - deduction;

        if (withdrawAmount > 50000) {
            balance = balance - (balance * 0.05);  // 5% tax
        }

        cout << "\nAccount Type: Savings";
    }

    else if (accountType == 'C' || accountType == 'c') {
        deduction = 100;
        balance = balance - withdrawAmount - deduction;

        if (withdrawAmount > 50000) {
            balance = balance - (balance * 0.05);
        }

        cout << "\nAccount Type: Current";
    }

    else {
        cout << "Invalid account type!";
        return 0;
    }

    cout << "\nAccount Number: " << accountNumber;
    cout << "\nWithdrawn Amount: " << withdrawAmount;
    cout << "\nRemaining Balance: " << balance;

    return 0;
}
```

OUTPUTS

1

```
Enter your Rollno and Marks obtained in five subjects: 192
480
Total Marks in five subjects: 480
percentage is: 96%
```

2

```
Enter the value of a b c:
23 45 6
23 45 6
45 6 23
```

3

```
Enter temp in fahrenheit:
23
The temp in celsius is -5
```

```
Enter elements of first 3x3 matrix
2
3
4
5
6
7
7
7
8
Enter elements of second 3x3 matrix
3
3
4
23
4
5
443
5
9
====Sum of Matrices====
5 6 8
28 10 12
450 12 17
====Product of matrices====
1847 38 59
3254 74 113
3726 89 135
```

5

```
Enter the radius of sphere:
23
Surface Area of Sphere = 6644.24
Volume of Sphere = 50939.2
```

6

```
Enter account type (S for Savings, C for Current): s
Enter account number: 4567
Enter amount to withdraw: 1000000
Error: Cannot withdraw more than 100,000 at a time.
```